BEYOND EVALUATION USE

An Empirical Study of Involvement and Use/Influence in Multi-Site Evaluations

Report 12:
A Case Study of the Impact of the
Robert Noyce Scholarship Program Evaluation

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Introduction

In November 2004, the National Science Foundation (NSF) awarded Dr. Frances Lawrenz and Dr. Jean King of the University of Minnesota a three-year grant to study the effect that participation in program evaluations has on the subsequent use of evaluation results. This grant, entitled "Beyond Evaluation Use", examined multiple projects within each of four main NSF-funded programs. These programs included: Local Systemic Change (LSC), Collaboratives for Excellence in Teacher Preparation (CETP), Advanced Technological Education (ATE), and Mathematics and Science Partnerships (MSP).

The grant activities were aimed at measuring and describing how these four programs carried out their multi-site evaluations. In particular, the research question was how these evaluations involved the various projects in this process. A further question focused on whether involvement affected the ways in which projects, individuals involved with the projects, and the STEM education and evaluation communities used the results.

As part of the research project, an on-line survey addressing the program-level evaluations was administered to evaluators and project leaders. The purpose of that *first* survey was to investigate the interaction of use and involvement and the data from that survey has been reported in "Report 1: NSF Follow-up Survey Detailed Data Report." Detailed case studies for each of the four programs (LSC, CEPT, ATE and MSP-RETA), which examine each program's history and evaluation context, and discuss survey findings within the context of each program can be found in Reports 2 – 5.[1]

After the administration of the *first* survey, the Beyond Evaluation Use team decided that further research into the relationship between evaluation use and influence should be done. The team re-administered the survey to gather more data for a confirmatory factor analysis. As with the *first* survey, the *second* survey was sent to PIs and co-PIs. This report summarizes the results of the *second* administration of the original survey of ATE, LSC, CEPT and MSP project leaders. The purpose of this second administration was to gather data for a confirmatory analysis. Since this data was available, however, a presentation and analysis of the results is being given here. As was done with the first survey, an analysis of the results is given to help explain the extent to which Noyce projects believed they were involved in the planning and implementation of the evaluation, and how that involvement affected the use of the program findings.

The primary research questions guiding this inquiry are as follows:

1. What is the relationship between involvement in the NOYCE program evaluation and the use and influence of the evaluation on project-level staff?
2. What factors other than involvement appear to be related to the use and influence of the program evaluation on the projects?

[1] http://www.cehd.umn.edu/projects/beu/ . See "Project Documents", Reports 2-5.

Methods

Data were collected through an online survey distributed to NOYCE project leaders. Additional data were collected through review of archival documents, publications, and reports produced by the NOYCE program and program evaluation.

NOYCE Project Leader Survey

The Beyond Evaluation Use team acquired names and email addresses for the principal investigators (PIs) and co-principal investigators (co-PIs) from the NSF website. Although the lists contained the most recent email addresses on record, some names were missing email addresses. In such instances, a member of the EUG team searched the Internet for an email address.

The PIs and co-PIs received an initial invitation email with a description of the survey, an explanation of its use, the Human Subjects approval number, and telephone numbers to call with questions. The email also contained a link to the online survey with a login and password. Two and five weeks following the first email, the team sent email reminders to non-respondents. The reminder email also contained a link to the online survey with a login and password. The data were collected from April 8, 2009 through May 31, 2009.

The second survey data is available in "Report 9: NSF Second Follow-up Survey Detailed Data Report". Survey findings presented in this report are taken from 57 close-ended items and four open-ended questions related to the involvement in and use of the NOYCE program evaluation. There was also an opportunity for respondents to provide additional comments in a final open-ended item

Due to the fact that many of the email addresses were obtained via Internet search by the EUG team, only approximately 10 percent of the emails were returned as undeliverable. After receiving the initial email invitation, a number of potential respondents replied with valid reasons for why they should not be included in the survey, such as those on projects that had not yet begun the evaluation process or those who retired prior to their project being evaluated. These persons were removed from the sample. In total, 25 of the original 222 individuals contacted were either removed from the survey or had emails returned as undeliverable. Of these, 96 individuals replied resulting in a 49% response rate.

Team members reviewed responses to open-ended items. When appropriate, responses may be used to support or exemplify findings from the close-ended survey items, which are described in the next section of this report. The following table shows the number and percentage of respondents who provided comments to each open-ended question or statement.

Table 1. Responses to open-ended items

Items 59, 60, 64	Comment	No Comment	Percent Commenting
What major factors, if any, enabled the evaluation experience to influence you?	28	68	29
What major factors, if any, limited the influence the evaluation experience had on you?	29	67	30

Is there anything else you would like to share with us about your experiences with the program evaluation?	23	73	24

Nonresponse study. To check for nonresponse bias, the team conducted a nonrespondent follow-up survey subsequent to the web survey of the PIs and co-PIs. Non-response error results when the people who responded to the survey are different from the individuals in the sample who did not respond, in a way that is relevant to the study. We developed a three-question nonrespondent survey to address the topics of involvement and use, as well as to gather some information about what kept individuals from responding to the initial survey and reminder emails. The nonrespondent survey items were:

1. How involved were you in the NOYCE Program Evaluation?
2. How much impact did the NOYCE Program Evaluation have on you?
3. What was the main thing that kept you from responding to our initial survey request?

A random sample of nonrespondents to the survey was drawn using Microsoft Excel. Each person was sent a personal, individually addressed email with a request for his or her input on the three items. Two reminder emails were sent to each of the respondents who did not respond to the initial request. Then, if email contact did not solicit a response, the individual was contacted by telephone. Sixty-three percent of the nonrespondents (22/35) completed the nonrespondent survey.

The results of the analysis comparing the NOYCE respondents and nonrespondents showed differences in the overall levels of involvement and use reported. The Involvement Mean of the survey respondents was higher (1.99) than the mean of the non-respondents (1.36). Similarly, the Use/Impact Mean was higher (1.98) for respondents than for non-respondents (1.32). The Involvement Mean for the respondents was calculated as the mean of thirteen questions on the original survey. The Use/Impact Mean was calculated in the same way, using the fourteen survey questions related to evaluation use and impact. The means for the non-respondents were simply the means of their answers to the two questions they responded to via email. Though these measurements are not identical, the two sets of means closely mirror one another in that the 4-point scale for each question or set of questions is the same.

The process of distilling the larger survey's multiple items related to use into a single question resulted in the choice of the word "impact" as a global concept to encompass questions about use of the process, instruments, and results of the evaluation. Therefore, the difference between respondents and nonrespondents may indicate the possibility of upward bias in the estimates of overall use for NOYCE survey respondents or it may also be an artifact of the difference between the terms use, influence and overall impact.

NOYCE Program Facts

Period: 2002 - present

Purpose: To encourage talented students in science, technology, engineering and mathematics to become K-12 mathematics and science teachers.

Primary Target: Institutions of higher education that provide scholarships and stipends to supporting students committed to teaching in high need K-12 schools.

Archival Review

The collection of archival documents for NOYCE was conducted primarily using electronic databases and search engines. The NSF website provided information on the NOYCE program as a whole, as well as on individual NOYCE projects. The archival documents included reports, grant products, survey instruments, journal articles, NSF grant solicitations, project summaries, and newsletters.

NOYCE Program and Evaluation Background and Context

NOYCE Program

The National Science Foundation's (NSF) Robert Noyce Teacher Scholarship Program was funded to increase the number of highly qualified teachers working in high need schools. This program has individual projects situated nationally at various university sites. The overall purpose of these projects is to recruit and support individuals with a strong academic background in science, technology, engineering and mathematics (STEM) content areas in becoming teachers. In return, recipients of scholarships and stipends must commit to serving two years as a mathematics or science teacher in a high need local educational agency.

NOYCE Evaluation Facts

NSF Grant Funding
- $147 million

NSF Goals for NOYCE Program Evaluation
- Provide evidence of quality of NOYCE program
- Contribute to the knowledge base of effective strategies for attracting and retaining effective teachers with strong STEM content knowledge.

Evaluation Questions
- To what degree is the program achieving its goals?
- Is it making an impact and reaching the individuals and groups intended?
- How effective is it when it reaches its constituents?
- Are there ways to significantly improve the program?

NOYCE Program Timeline and Budget

Between 2003 and 2009, NOYCE funded 207 grant proposals totaling approximately $147 million. Funding support was awarded to colleges and universities throughout the United States primarily for use in undergraduate programs. A small number of these awards (approximately 7%) went to master's degree programs. As per NSF requirements, NOYCE awards went to support scholarships, stipends, and programs. Grantees frequently established partnerships, most commonly with local school districts or departments and programs within an institution, but also with other institutions of higher education, and discipline-based organizations such as the Physics Teacher Education Coalition. Awards ranged from approximately $75,00 to $1.5 million, with an average award of just over $700,000.

NOYCE Program Evaluation Background

To evaluate the effectiveness of NOYCE on various dimensions, the University of Minnesota Noyce Program evaluation project was funded in 2005.The evaluation team has been composed of a number of graduate students and postdoctoral students with Frances Lawrenz as the principal investigator (PI).

NOYCE Program Evaluation Questions

The Noyce program evaluation assessed the impact and effectiveness of NOYCE by addressing three basic evaluation questions[2]:

1. What are the characteristics of the teacher education programs provided by the Noyce projects?
2. In the opinions of the PIs and scholars, what role did the Noyce funding play in the scholars' (i.e. Noyce scholarship/stipend recipients) decisions to teach in high need schools?
3. In the opinions of PIs, scholars, STEM faculty and districts in what ways and to what extent are the characteristics of the Noyce program related to
    - Recruiting individuals with strong STEM backgrounds, including those who might not otherwise consider teaching/going into teaching careers
    - Retaining scholarship/stipend recipients in the teaching profession
    - Developing the pedagogical content knowledge and ability of scholarship/stipend recipients to teach STEM content
    - Developing positive profiles for pre-service teachers in STEM discipline departments

NOYCE Program Evaluation Design

The central components of the participatory approach of the U of MN Noyce evaluation are the evaluation question specification and data collection efforts collaboratively designed by the U of MN Noyce evaluation team and the projects.[3] By including representatives from each of the Noyce projects, the multi-site participatory program evaluation hoped to incorporate issues and concerns of the projects as well as the needs of NSF. The information from the literature review, the project evaluation resources, and the existing monitoring data were used to help frame the participation. The hope was that the participatory approach would address the problems associated with many multi-site evaluations, such as substantial variability in implementation, not enough information on actual practice, lack of common and appropriate outcome measures, and the need to synthesize results for evaluating large-scale multi-site educational reform programs.[4] Additionally, this representational approach provides a highly skilled group of evaluators to help plan the overall evaluation thereby optimizing its quality.[5]

NOYCE Program Evaluation Components

There were several principal evalation activities that were carried out by the evaluation team. These included:

[2] "University of Minnesota Evaluation of the Robert Noyce Teacher Scholarship Program, Final Report Section Six: A Model of the Pathway to Retention in High Need Settings, Analysis of the Noyce Scholar Interview", from http://www.cehd.umn.edu/EdPsych/NOYCE/Reports/default.html
[3] Lawrenz & Huffman, 2003
[4] Hamilton, McCaffrey, Stecher, Klein, Robyn & Bugliari, 2003
[5] Leff & Mulkern, 2002

- *Literature Review*. The U of MN Noyce evaluation team conducted an extensive review of the literature available on STEM K-12 teacher recruitment and retention. The literature database is posted on the project website.[6]
- *Thematic Synthesis of the Existing Evaluation Information.* The evaluation team also conducted a thematic synthesis of the existing evaluation information. The database is posted at the program evaluation Web site. Web links to evaluation instruments being used by specific projects are also available through the Web site.
- *Statistical Query of Existing Monitoring Data.*. A statistical query of the existing monitoring data was also performed.
- *Surveys*. Web-based grantee surveys formed the backbone of the NOYCE evaluation. Surveys were developed for PIs, scholars, and STEM faculty. The evaluation questions and instruments for these surveys were created jointly by the evaluation team and the NOYCE PIs through a series of meetings and discussions.
- *Interviews.* Semi-structured interview protocols were developed for school district representatives and scholars. The interviews generally were between 30-50 minutes and were recorded and transcribed verbatim for analysis.
- *Video conferences and presentations*. Video conferences and presentations at the NSF PI meetings were also held. Topics of these meetings include, but were not limited to how to conduct the evaluation, suggestions for analyses and how information should be disseminated.

Project Leaders Involvement and Use

This section presents data from the survey to describe how project leaders (i.e., PIs and co-PIs) reported being involved in and influenced by the Noyce program evaluation. This presentation of findings is organized around six topics that guided the data collection and analysis for the study. For each topic, quantitative results are discussed. The six topics are:

- Motivation for Involvement
- Levels of Involvement
- Perceived Effect of Involvement
- Use by Project Leaders and Evaluators
- Use by Projects
- Factors Affecting Use

Motivation for Involvement

Item 4 of the survey addressed the motivation that PIs and co-PIs had, at different stages (planning, implementation and communication) of the evaluation, for being involved. The planning stage includes activities such as discussing the focus of the evaluation, identifying evaluation planning team members, developing the evaluation plan, while the implementation stage includes developing data collection instruments and processes, collecting, analyzing, reviewing, and interpreting data. The final stage of the program evaluation – the communication stage – is comprised of activities such as developing future plans for your project, writing and

[6] http://www.cehd.umn.edu/EdPsych/NOYCE/default.html

reviewing evaluation reports, presenting evaluation findings. Of the three stages of the evaluation, findings suggest that motivation for involvement was predominantly due to being either required or asked, with being required greatest at the communication stage (38.8%) and being asked greatest at the planning stage (43.8%). While volunteering was not the most frequent response for any stage, it ranged from 24.6% of respondents for the implementation stage to 34.7% for the communication stage.

Table 2 *Motivation for Involvement*

Item 4	**I was required**	**I was asked**	**I volunteered**	**N**
Planning stage	25.0%	43.8%	31.3%	64
Implementation stage	36.2%	39.1%	24.6%	69
Communication of evaluation findings stage	38.8%	26.5%	34.7%	49

Levels of Involvement

The survey asked respondents about their perceptions of involvement during the planning, implementation and communication phases of the program evaluation. Responses to the involvement questions are summarized in Tables 3, 4 and 5. Each involvement question began with "I was involved in…."

Table *3 Involvement - planning*

Items 1 - 3	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
Discussions that focused the evaluation.	27.7%	23.4%	23.4%	25.5%	94	2.47	1.15
Identifying evaluation planning team members.	48.9%	9.6%	16.0%	25.5%	94	2.18	1.29
Developing the evaluation plan.	41.9%	18.3%	17.2%	22.6%	93	2.20	1.21

Table 4 *Involvement - Implementation*

Items 5 - 10	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
Developing data collection instruments.	59.1%	15.1%	12.9%	12.9%	93	1.80	1.10
Developing data collection processes.	54.8%	18.3%	10.8%	16.1%	93	1.88	1.14
Collecting data.	47.3%	20.9%	7.7%	24.2%	91	2.09	1.24

Reviewing collected data for accuracy and/or completeness.	49.5%	16.5%	16.5%	17.6%	91	2.02	1.17
Analyzing data.	66.3%	9.8%	9.8%	14.1%	92	1.72	1.12
Interpreting collected data.	52.2%	15.2%	16.3%	16.3%	92	1.97	1.16

Table 5 *Involvement - Communication*

Items 12 - 15	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
Writing evaluation reports.	70.0%	6.7%	11.1%	12.2%	90	1.66	1.09
Reviewing evaluation reports for accuracy and/or completeness.	50.0%	18.9%	13.3%	17.8%	90	1.99	1.17
Presenting evaluation findings.	67.4%	5.6%	5.6%	21.3%	89	1.81	1.25
Developing future project plans based on evaluation results.	52.7%	11.0%	9.9%	26.4%	91	2.10	1.30

For all three phases, the majority of respondents indicated either ‘no’ or only ‘a little’ involvement for each question. The most extensive overall involvement was in the planning stage. During this phase, more than a quarter of all respondents (25.5%) felt extensively involved in discussions that focused the evaluation and in identifying evaluation planning team members, while slightly fewer were extensively involved in developing the evaluation plan (22.6%). One explanation for this comes from one respondent’s comment that “I enjoy the evaluation process and would have liked to have been more involved in the implementation and dissemination; however, I was not the PI during the course of the project although I was part of the management team.”

One reason for these findings has to do with the types of opportunities that were made available to PIs and co-PIs. Their primary opportunities for involvement came in planning stage of the evaluation and in the development of data collection instruments. Program evaluators then presented the data from the evaluation implementation process to PIs, who were then afforded the opportunity to comment on these finding. Reports were written by the program evaluation team, but opportunity to respond to these reports was limited.

During the implementation stage, the most extensive involvement seen was in collecting data (24.2%). At one point during the data collection stage, all PIs were surveyed. While the greatest number of respondents reported no involvement in developing data collection instruments (59.1%) and analyzing data (66.3%), attempts were made, in the form of webinars, to involve project leaders in this process. Difficulties with the webinar appear to be primarily responsible for this lack of involvement.

During the communication phase of the evaluation, more than a quarter of all respondents (26.4%) perceived themselves as being extensively involved with the development of future projects plans based on the evaluation results. And while most respondents (70%) felt no involvement in writing the evaluation reports, they nevertheless did see themselves as somewhat more involved with reviewing evaluation reports and presenting evaluation findings. One reason for this may have been the presentations given at PI meetings where participants were involved in providing feedback on findings.

Overall, PIs and co-PIs characterize the evaluation planning stage as that stage of the evaluation where they were most extensively involved. Moving into the implementation stage saw an increase in the number of respondents perceiving themselves as less involved. The communication stage continued this trend somewhat.

Perceived Effect of Involvement

Questions 17 through 22 concerned respondents' perceptions concerning the influence their input had on each stage of the program evaluation. In providing this feedback, respondents were asked to differentiate between having their input taken into consideration and having their input actually used. Survey results suggest that PIs and co-PIs perceived their input as being both considered as well as used to some degree during each stage of the evaluation. That is to say that input use was not perceived differently from input consideration at any stage of the evaluation, suggesting that respondents' judged their input to be influential overall. Moreover, the degree to which they viewed their influence as either considered or used was also similar across the stages of the evaluation. Between 20-25% of respondents said that their input was both considered and used extensively in the evaluation, while another 14-18% claimed that their input was considered or used "some".

Table 6 *Influence on the Evaluation*

Items 17 - 22	Was not involved	No	Yes, a little	Yes, some	Yes, extensively	Do not know	N
Input during *planning* was taken into consideration.	29.5%	1.1%	17.9%	15.8%	25.3%	10.5%	95
Input during implementation was taken into consideration.	33.7%	.0%	11.6%	17.9%	24.2%	12.6%	95
Input during communication was taken into consideration.	44.7%	.0%	7.4%	14.9%	20.2%	12.8%	94
Input during planning was used.	30.5%	3.2%	15.8%	17.9%	20.0%	12.6%	95
Input during implementation was used.	35.8%	2.1%	15.8%	15.8%	20.0%	10.5%	95
Input during communication was used.	43.2%	2.1%	9.5%	13.7%	20.0%	11.6%	95

Perceived Use by Project Leaders

The survey asked project leaders to reflect on the extent to which the program evaluation influenced their knowledge/understanding, skills and beliefs. Additionally, project PIs were

asked about their use of the program evaluation process and findings in another evaluation context. Each of these is examined in turn.

Influence on Knowledge and Understanding

Respondents most commonly believed that their knowledge and understanding of evaluation (all stages), STEM education and evaluation, and their own project increased "a little" or "some" as a result of the program evaluation. While the program evaluation appears to have most strongly influenced respondents' knowledge and understanding of how to plan and implement an evaluation, communication was also influenced for the majority of project leaders. With respect to STEM education and evaluation, respondents reported that the program evaluation was less likely to increase knowledge and understanding of STEM education (M = 1.79, SD = 0.89) than STEM education evaluation (M = 1.95, SD = 0.88). Over 48% of respondents said that the program evaluation had no influence on their knowledge and understanding of STEM education. Finally, 60% of respondents believed their knowledge and understanding of their own projects was increased at least "a little" as a result of the program evaluation, with almost 14% reporting that the increase was extensive. Given the level of knowledge concerning STEM education, these results are not surprising.

Table 7 *Influence on Knowledge and Understanding*

Items 25 – 30	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
The evaluation increased my knowledge/understanding of how to plan an evaluation.	35.8%	32.6%	27.4%	4.2%	95	2.0	.90
The evaluation increased my knowledge/understanding of how to implement an evaluation.	35.8%	33.7%	27.4%	3.2%	95	1.98	.88
The evaluation increased my knowledge/understanding of how to communicate evaluation findings.	43.2%	24.2%	30.5%	2.1%	95	1.92	.91
The evaluation increased my knowledge/understanding of STEM education.	48.4%	28.4%	18.9%	4.2%	95	1.79	.89
The evaluation increased my knowledge/understanding of STEM education evaluation.	36.8%	35.8%	23.2%	4.2%	95	1.95	.88
The evaluation increased my knowledge/understanding of my project.	26.3%	27.4%	32.6%	13.7%	95	2.34	1.02

Influence on Skills

Although the program evaluation's influence on skills reflects a similar pattern to its influence on knowledge and understanding, slightly higher percentages of respondents reported no influence on skills (see Table 8). One inference is that it is easier to affect understanding than skills, especially where the level of contact is low. The pattern was also similar to those found for influence on knowledge and understanding in that STEM education was the area with the least amount of influence as indicated by lowest mean and highest percentage reporting no influence. Further, respondents reported that their evaluation planning skills were those most influenced in all of the stages of evaluation.

Table 8 *Influence on Skills*

Items 31 - 36	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
The evaluation improved my skills in planning an evaluation.	42.1%	28.4%	21.1%	8.4%	95	1.96	.99
The evaluation improved my skills in implementing an evaluation.	47.4%	24.2%	21.1%	7.4%	95	1.88	.99
The evaluation improved my skills in communicating evaluation findings.	51.6%	23.2%	20.0%	5.3%	95	1.79	.94
The evaluation improved my skills as a STEM educator.	52.6%	26.3%	18.9%	2.1%	95	1.71	.85
The evaluation improved my skills as a STEM education evaluator.	44.2%	31.6%	20.0%	4.2%	95	1.84	.89
The evaluation improved my skills for working on my project.	33.0%	38.3%	21.3%	7.4%	94	2.03	.92

Influence on Beliefs

In much the way that respondents were asked to reflect upon the program evaluation's influence on their knowledge, understanding and skills, respondents were asked to further reflect upon the influence that the evaluation had on their *beliefs* about the importance of the evaluation. Table 9 presents these results. This time we see a pattern that differs significantly from the influence respondents reported for their knowledge and understanding as well as for their skills. The difference is the significant increase in the number of respondents that report extensive influence, and the corresponding drop in no or little influence. One further observation is the relatively larger influence on respondents' beliefs about the importance of program evaluation than on either STEM education or STEM education evaluation, or the importance of an individual's project. This can be explained in terms of the fact that many PIs come to the evaluation having a high level of belief in the value of STEM while their knowledge of evaluation is relatively limited. Hence, beliefs about evaluation might be easier to change in these instances.

Table 9 *Influence on Beliefs*

Items 38 - 43	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
The evaluation increased my belief in the importance of planning an evaluation.	37.2%	26.6%	17.0%	19.1%	94	2.18	1.14
The evaluation increased my belief in the importance of implementing an evaluation.	38.6%	30.1%	19.4%	14.0%	93	2.11	1.06
The evaluation increased my belief in the importance of communicating evaluation findings.	38.3%	29.8%	18.1%	13.8%	94	2.07	1.06
The evaluation increased my belief in the importance of STEM education.	48.9%	21.3%	18.1%	11.7%	94	1.93	1.07
The evaluation increased my belief in the importance of STEM education evaluation.	44.7%	23.4%	20.2%	17.0%	94	1.99	1.06
The evaluation increased my belief in the importance of my project.	37.2%	25.5%	20.2%	17.0%	94	2.17	1.11

With respect to questions 25-43, all of which concern the same sort of influence that the evaluation had, it is important to note that this influence may have been a negative influence. For example, the program evaluation may have failed to give adequate advance notification for the completion of a task and, as a result, the respondent learned that it was important to do it differently. To address issues of this sort, question 37 asked whether the evaluation had a negative impact and then offered space for a written explanation. Only 7% of survey participants responded to this question (7/96). With so few responses, there were no trends identified.

Nevertheless, a couple of comments alluded to time concerns. One respondent wrote that "[the evaluation] is an important part of any activity and one that is worth doing. It just takes time." Another wrote "all of the co-PIs who were supposed to design and implement the evaluation either left the project or refused to do any work. Everything was left to me, yet that was never to be my role in the project…"

One respondent expressed concern about the negative impact that a focus on evaluation process may be having on the goal of understanding outcomes. In articulating this concern, this respondent questioned the role of outside evaluators. "I've become concerned about the cost-benefit ration of an evaluation system that relies on "professionals" who have such a strong a vested interest in the process rather than on simple collection of key outcomes."

Use in another Context

Questions 45 – 49 sought information from respondents concerning the use of the evaluation knowledge, products, or processes in another (subsequent) evaluation context. The survey directed only respondents who had participated in a subsequent evaluation to respond to these questions. If a respondent had not participated in another evaluation, they were instructed to continue with question 50.

For those respondents who participated in another evaluation after the NOYCE program evaluations (39 out of the 96 respondents), almost 60% used what they learned from planning or from implementing the evaluation in a subsequent evaluation context. "The evaluation process used in this project has helped me in designing and developing other evaluation strategies for other projects," noted one respondent.

Although they reported a high rate of using what they learned from these two stages, respondents, nevertheless, were much less likely to use the actual evaluation plan from the program evaluation. Over 60% of respondents said they did not use the plan at all as a model for a future evaluation. For the least used aspect of the evaluation, the data collection instruments (with 79.5% reporting that they did not use the data collection instruments in another evaluation), there was over 20% of respondents using those instruments at least "a little" in another evaluation subsequent to the NOYCE program evaluation.

Table 10 *Use of Knowledge, Products, Processes in Another Evaluation Context*

Items 45 - 49	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
I used what I learned from planning the NOYCE program evaluation in another evaluation.	41.0%	28.2%	25.6%	5.1%	39	1.95	.94

I used the evaluation plan from the NOYCE program evaluation as a model in another evaluation.	61.5%	23.1%	12.8%	2.6%	39	1.56	.82
I used what I learned from implementing the NOYCE program evaluation in another evaluation.	41.0%	33.3%	23.1%	2.6%	39	1.87	.86
I used data collection instruments from the NOYCE program evaluation in another evaluation.	79.5%	5.1%	10.3%	5.1%	39	1.41	.88
I used what I learned from communicating evaluation findings from the NOYCE program evaluation in another evaluation.	61.5%	12.8%	23.1%	2.6%	39	1.67	.93

Use of Findings

Questions 50 – 58 asked PIs and co-PIs to report the ways in which they used the program evaluation findings. Table 11 reports these results. One notable outcome of this series of questions is the large percentage of respondents using the program evaluation findings to develop future plans for their projects (20.7% used them "extensively", 25.0% using them "some", and 29.3% used them at least "a little"). Respondents also used the program evaluation finding to meet contractual or legal requirements (M = 2.29, SD = 1.25) and to make decisions about the future existence of their projects (M = 2.19, SD = 1.13). On the other hand, using the program evaluation finding to promote additional evaluation of their projects was the least likely use of findings that respondents reported. Over 58% of respondents claimed that they did not use evaluation findings for this purpose.

Table 11 *Use of Findings*

Items 50 - 58	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
I used the program evaluation findings to make decisions about the future existence of my project (e.g., cancel or continue).	36.7%	26.7%	17.8%	18.9%	90	2.19	1.13
I used the program evaluation findings to develop future plans for my project.	25.0%	29.3%	25.0%	20.7%	92	2.41	1.08
I used the program evaluation findings to increase the attention given to my project.	39.6%	20.9%	25.3%	14.3%	91	2.14	1.10
I used the program evaluation findings to meet contractual and/or legal requirements of my project.	41.3%	14.1%	18.5%	26.1%	92	2.29	1.25
I used the program evaluation findings to enhance my organization's commitment to my project.	47.8%	14.1%	17.4%	20.7%	92	2.11	1.22
I used the program evaluation findings to seek funding for my project(s).	52.2%	17.4%	14.1%	16.3%	92	1.95	1.15
I used the program evaluation findings to promote additional evaluation of my project(s).	58.2%	19.8%	12.1%	9.9%	91	1.74	1.02
I used the program evaluation findings to make changes to my project(s).	35.9%	32.6%	18.5%	13.0%	92	2.09	1.03
I used the program evaluation findings to help plan a new project(s).	51.1%	19.6%	17.4%	12.0%	92	1.90	1.08

Comparisons Among the Types of Influence

As shown in Table 12, data from the survey suggest that the NOYCE program evaluation had a somewhat low level of influence on project leaders across all categories. The greatest influence can be seen on project leaders' ways of using findings (M = 2.09, SD = 0.92) and beliefs (M = 2.08, SD = 0.95). The only area where influence lagged (M = 1.69, SD = 0.67) was the subsequent use of what was learned from the program evaluation. Here, however, we must be careful interpreting results due to the small number of respondents (N = 39) in comparison with the other areas of influence.

Table 12 *Influence of Program Evaluation on Project Leaders*

Type of Influence	**N**	**Mean**	**SD**
Influence on knowledge/understanding (q25-30)	95	1.99	.78
Influence on skills (q31-36)	94	1.87	.82
Influence on beliefs (q38-43)	93	2.08	.95
Use of knowledge, products, processes in other contexts (q45-49)	39	1.69	.67
Ways of using findings (q50-58)	91	2.09	.92

Use by Projects

As we saw above, survey items focused on the ways in which project leaders were influenced by the program evaluation, including the extent to which they used the process and findings to make decisions about projects. Some of the open-ended responses allowed for deeper understanding about the ways in which the program evaluation influenced the projects. Open-ended items are listed in Table 1, along with the number of respondents commenting and the rate of commenting.) One important caveat concerning the open-ended items concerns the fact that some of the respondents answered these questions in reference to their own project's evaluation, as opposed to the larger program evaluation.

Nevertheless, evaluation team members assessed the open-ended items to identify those aimed at the program level evaluation. One theme that emerged concerned the role of evaluation findings and their availability as well as summary presentations of results. One respondent wrote:

> For the most part, the fact that enabled some influence of the University of Minnesota R3 evaluation data on me was the summary presentation of results at the annual Noyce conferences and the existence of the websit where evaluation information is made available.

This sentiment was echoed by a number another respondent who gave a more specific account of the role of these finding at the local level. "The report of evaluation findings was useful particularly in how it confirmed impressions and local evaluation data.". Reporting and presenting findings of the program evaluation was seen as both infomative and useful at the project level.

Influence on the Field

Questions 61 and 62 looked at respondents' assessment of how influential the NOYCE program evaluation was on the STEM education community and the evaluation community. Tables 13 and 14 summarize these survey responses.

Table 13 *Influence on STEM Education Community*

Not influential at all	Marginally influential	Somewhat influential	Influential	Very influential	Total
14.3%	29.8%	34.5%	13.1%	8.3%	84

Table 14 *Influence on the Evaluation Field*

Not influential at all	Marginally influential	Somewhat influential	Influential	Very influential	Total
17.7%	46.8%	26.6%	6.3%	2.5%	79

While a larger percentage of respondents believed that the evaluation was "not influential at all" or "marginally influential" on the field of evaluation (64.5%) than on the STEM community (44.1%), in general respondents seemed to have comparable attitudes towards these two targets of evaluation influence. If anything can be noted in these responses it is only that the influence on the evaluation field is viewed as significantly less than on the STEM community.

Factors Affecting Use

Survey respondents were asked to identify any major factors that either enabled or limited the influence of the evaluation on them. We have already noted the influence of the availability of summary presentations and findings of results for the project-level . This factor was also noted as being more generally influential on respondents. As for limiting factors, the generality of the program evaluation, along with the lack of focus that respondents perceived as inherent to the program evaluation were commonly named.

Enabling Factors

Numerous respondents commented on the positive influence of the electronic availability and access to information concerning the NOYCE project evaluations. The "ease of access to evaluation reports"was noted by one respondent, and another specified "the existence of the website where evaluation information is made available". A third respondent noted "reading reports, which are submitted in a timely manner" helped to enable the influence of the program evaluation. Even in cases where the evaluation was not deemed particularly influential, the published results were noted as being informative. A fourth respondent claimed:

> Other than learning about the results of the evaluations by the University of Minnesota R3 project after they had been reported (which was informative, but did not significantly shape the way we ran our Noyce program at our univeristy), I was not influenced by the

> evaluation process, because I was not involved in the planning, design, implementation or reporting activities

Limiting Factors

Among respondents, there seemed to exist some concern about the purpose of the evaluation. In particular, the purpose of the evaluation or exactly what respondents were supposed to be considering as "the evaluation" under consideration, and the generality of the evaluation were both called into question. One respondent questioned claimed "The Noyce evaluation program was difficult and sometimes unclear. I spent a signficant amount of time trying to figure out what to do to proceed when responding to most of the questions." This concern was mirrored by another respondent who claimed that

> I think the Noyce program is very difficult to adequately evaluated on a national scale. Independent project evaluations are more useful. At the antional level, the evlaution should simply be the reporting of numbers of scholars and where they end up teaching (not really an evalautions – more of a collating of information).

One respondent who pointed out "this survey needs `does not apply' options for each question" perhaps summarized these frustrations best.

Confusion over what was being evaluated was a limiting factor. The generality of the evaluation was also mentioned.

> I found the overall program evaluation to be too general to influence specific aspects of our project. The specific information that was returned to us was no more useful than our local evaluation.

These limitations – the aim of the evaluation and the generality – were the primary limiting factors that respondents identified.

Summary

This document summarizes the ways in which NOYCE project leadership experienced involvement in the NOYCE program evaluation and how they subsequently used the results of this evaluation.

Of the three stages of the evaluation, findings suggest that motivation for involvement was predominantly due to being either required or asked, with being required greatest at the communication stage and being asked greatest at the planning stage. While volunteering was not the most frequent response for any stage, it ranged from one quarter of respondents for the implementation stage to over a third for the communication stage. Overall, PIs and co-PIs characterize the evaluation planning stage as where they were most extensively involved; more respondents perceived themselves as less involved at the implementation and communication stages. However, less than half of respondents perceived their input as being extensively or somewhat considered and used during each stage of the evaluation.

Respondents most commonly believed that their knowledge and understanding of evaluation (all stages), STEM education and evaluation, and their own project increased “a little” or “some” as a result of the program evaluation. Although the program evaluation’s influence on skills reflects a similar pattern to its influence on knowledge and understanding, slightly higher percentages of respondents reported no influence on skills. Significantly more respondents reported extensive influence on their beliefs, with a corresponding drop in no or little influence. For those respondents who participated in another evaluation after the NOYCE program evaluations, the large percentage of respondents reported using the program evaluation findings to develop future plans for their projects. On the other hand, they were least likley to use the program evaluation findings to promote additional evaluation of their projects.

Project leaders identified the role of the evaluator in timely data collection and reporting as having a positive influence on the projects. In addition, numerous respondents commented on the positive influence of the electronic availability and access to information concerning the NOYCE program evaluations. Among respondents, there seemed to exist concerns over the purpose of the evaluation and whether the generality of the evaluation reduced relevance to specific programs.